UTS

Australian Ethical Investment Ltd. – Report

Data visualisation & Visual analytics

Spring 2025

Assessment 1

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Introduction

Australian Ethical Investment Limited (AEF) is a fund management company that specializes in environmental and socially responsible investments. The company offers ethical managed funds (registered managed investment schemes) and Retail Superannuation Fund, Managed Fund and ETF.

Currently, the market value of the company's equity capital is around AUD $923 millions, and it has around 113.4 million shares quoted.

Initial exploration

The dataset used in this report, was sourced from DatAnalysis, in August 2025, compiling data between the 1st of July 2024 and the 30th of June 2025.

The original dataset includes the following variables:

- ASX code: categorical nominal qualitative variable that indicates the ASX of the company. This column was hidden because it was redundant for the objective of the assignment.
- Company name: categorical nominal qualitative variable that indicates the ASX of the company. This column was hidden because it was redundant for the objective of the assignment.
- Date: quantitative interval-scale variable that provides the date of when the information of the observation was collected.
- Open: quantitative ratio-scale variable, provides the stock price at the beginning of the trading session for that date.
- High: quantitative ratio-scale variable, provides the highest stock price reached during the trading session.
- Low: quantitative ratio-scale variable, provides the lowest stock price reached during the trading session.
- Close: quantitative ratio-scale variable, provides the final stock price at the end of the trading session for that date.
- Volume: quantitative ratio-scale variable, that shows the total number of shares traded during that day.
- Market capitalisation ($m): this quantitative ratio-scale variable provides the company's size and investor valuation (it is calculated by multiplying the share price at close by the number of shares on issue).
- Shares on issue: this quantitative ratio-scale variable, provides the total number or ordinary shares that have been issues by the company and are currently held by shareholders.

For this assignment, the data must be sorted by date, showing the oldest information first, and the most recent data, last. For that, I created a new column called "Timeseries", using the formula DATEVALUE, to provide a number assigned for each date, allowing me to sorter the data.

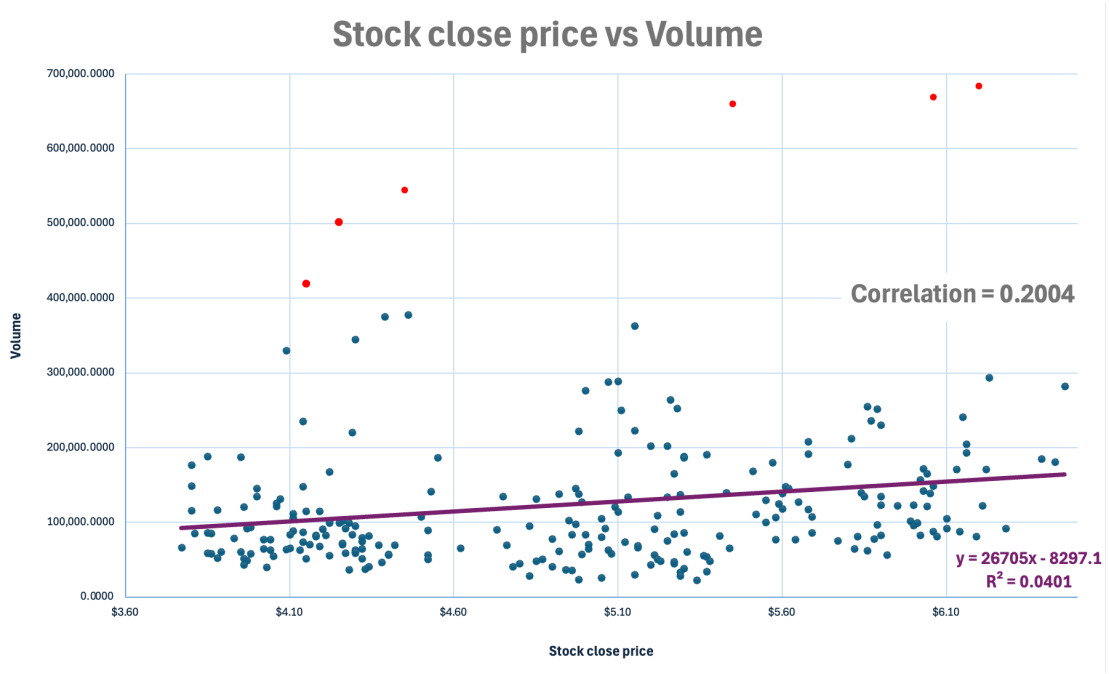
I also added format to each column, columns A (ASX code) and B (company name) are text, column C (date) is formatted as long-date, columns D (open), E (high), F (low) and G (close)

are price variables, as well as column I (market capitalisation in $m). Finally, columns H (volume) and J (shares on issue), are formatted as numbers.

Summary of dataset characteristics
Time frame: from 01/07/2024 to 30/06/2025, daily data
Number of variables: 10 (originally)
Number of observations: 253

Data visualisation and analysis

Stock close price vs volume analysis
For this visualization, I created a scatter plot chart, in order to identify if both variables follow a pattern of correlation or not. I used as $3.5 minimum value of close price, providing a closer look of each observation (dot). I provided axis labels for a deeper visual understanding. I also marked outliers in red, so it's easier to identify them.
To improve the visualisation, I rescaled the Y-axis and added a linear trend line with its formula.
I also created a new variable within the "Stock close price vs volume analysis" sheet, called Corellation (column A), that calculated the correlation between the close price and volume, for the overall period. Then, I added this information to the chart.



This chart doesn't show an existing linear correlation or strong and clear pattern between the stock close price and volume, and its coefficient (approximately 0.2, is low). This means,

there's no evidence of a trend where a stable increase of the stock close price is always related to a stable increase of volume, or backwards. The dots are very dispersed.
However, we can find specific zones of interest. For example, most of the dots (reach one representing an observation), are located in the bottom part of the chart, showing that most of the time, the volume stays relatively low. This suggests that the typical or "normal" trading activity happens with moderated volumes.
On the other hand, we can see some outliers within this chart, and all of them are located in the top part of the chart, between the volumes 400,000,000 and 700,000,000.
Observing this, we can infer that volume is not linearly correlated with the stock close price and that high volume peaks are unusual and don't represent a trend. Usually, this kind of peaks could be attributed to company's announcements or market events (such as merges or acquisitions).

Stock movement analysis

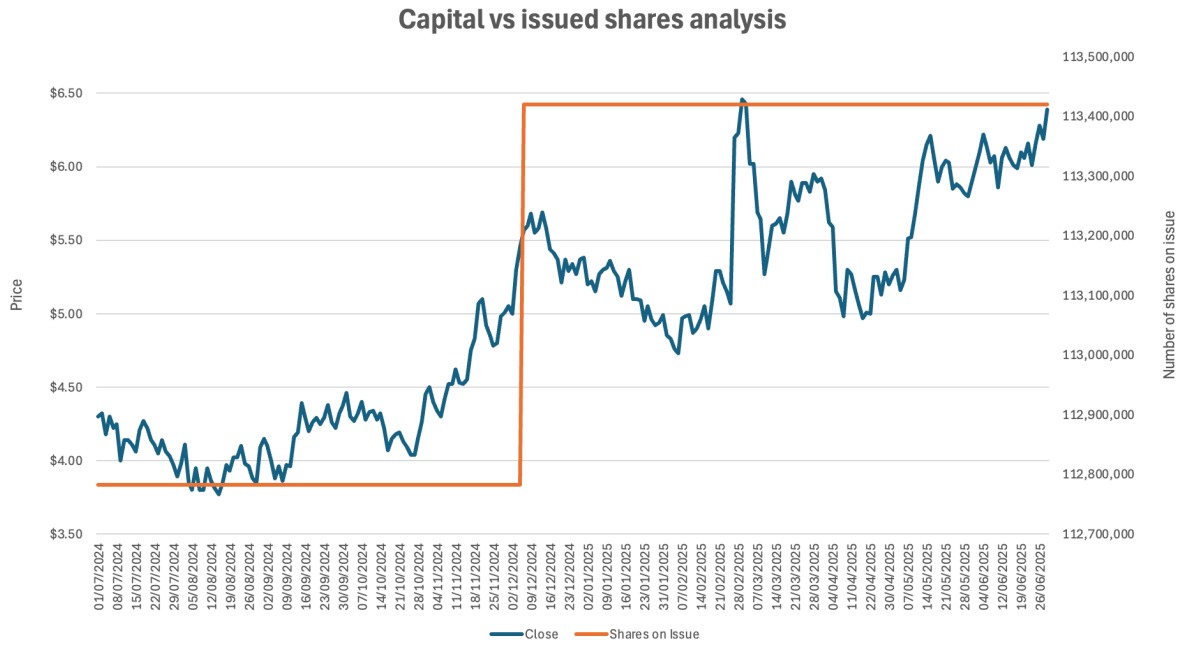
To analyse the variation of a stock price (open, high, low, close) and its timeframe, I used a candlestick chart, because this type of chart is specially created for this type of analysis. It shows accurately four data points in just one visualization. I used $3.5 as minimum and $6.75 as maximum value of prices (open, high, low, close), so we can have a closet look of the chart. I also applied a colour code on this chart; when the closing price is higher than the opening price (up bar), the candle is green, and when the closing price is lower than the opening price (down bar), the candle is red. This helps identifying how many greens and reds we see, giving further information on pricing over time. I removed the automatic legend, because it doesn't add relevant information.

This graph shows four periods of different trends. The first period, from July 2024 to late October 2024, we see a downward trend, because the stock price starts around $4.50 and shows a decreasing trend over that period. We can see more red candles than green. This suggests that the selling pressure is stronger.
The second period, from late October 2024 to December 2024 features transition and reversal. After reaching a minimum in October 2024, the price shows a reversal trend. The green candles start being more frequent and larger, beating the red candles. Now the trend begins

showing a gradual rise, now suggesting that the buying pressure is stronger. The price suggests a bullish market until around $5.00, then, the trend becomes more stable.
Next, we see the third period, from December 2024 to late February 2025. During this period, the price doesn't change much, and we can't really identify a trend or pattern. We see almost same frequency of green and red candles. However, the wicks length tends to vary often, showing indecision and volatility.
Last, the fourth period, between late February 2025 and late June 2025, show a strong bullish trend. The price breaks out and shows an upward trend. Also, during this period, green candlesticks are predominant, with longer bodies indicating a strong buying pressure. The stock reaches a new peak, over $6.00 with a high of approximately $6.50 in late May, before slightly stabilizing by June.

Capital vs issued shares analysis

As we want to find out the impact of issued shares on capital (if any), I used a line graphic, to check if both variables follow a pattern of correlation or not, over time. This way, we can see when the company issued or rebought stocks, and by putting issues shares and close price in the same chart, we can visually see how the price line varies according to the number of shares on issue. As the scales of both variables are very different, I created two vertical axes within the same chart.

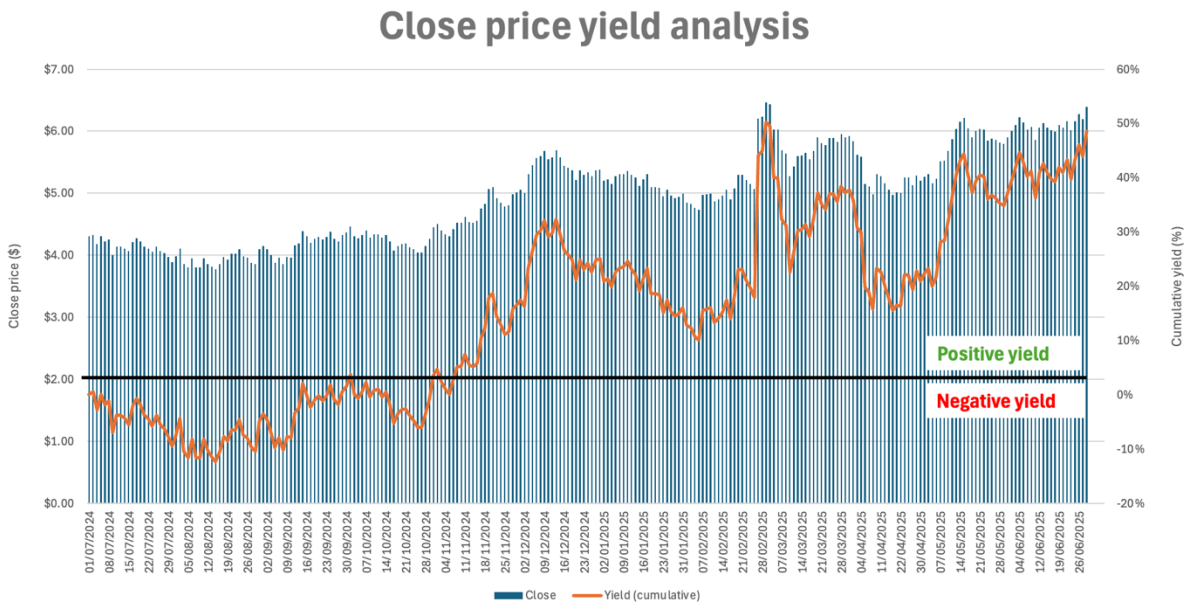


From this line chart, we can see three main phases. First, we see a period between July 2024 and November 2024. The number of issues shares is stable and relatively low, indicating that the company didn't issue or rebought stock significantly during that period. At the same time, the close price shows a downward trend, decreasing from around $4.50 to approximately $3.80. From this information, we can infer that the company slightly lost value during that period.

On the second phase, on late November 2024, the number of issues shares spiked to around 114,000,000 shares. We can also easily spot the market reaction, the close price doesn't decrease but starts an upward trend, increasing rapidly from its lowest value to around $5.50 per share. This means that the market interpreted the shares issue as a positive sign, showing an increased share price.
The third and last phase, from December 2024 to June 2025, shows a stabilization on shares on issue, and stays constant for the rest of the period. The close price line continues with an upward trend, reaching new peaks around $6.00. Even though we can see some lower close prices, the fact that the trend is still upward, means that the investors and the market values the company positively, even more after the shares issue.

Close price yield analysis

The yield is an important parameter that values the stock price vs dividend. In this instance, I didn't have access to the dividend payments, so instead, I used the close price data for this analysis. In this case, I will use a combo chart with secondary axis, to show the percentual variation of the close price over time. Using a percentual variation chart let see a normalized data series, as the start point goes from 0% to 100%, over time. By visualizing the data with this chart, it is easier to understand the accumulated profit/loss over time. I also added a line on the secondary axis, representing the 0% level, to provide a better understanding on positive and negative yield, as well as text boxes to label those levels.
For this analysis, I created a new variable called "Yield (cumulative), on column L, that uses the following formula: ([Close price$_k$ / Close price$_{k-1}$] – 1) and then formatted as percentage. In this formula, we consider k=day, replacing by every daily value.

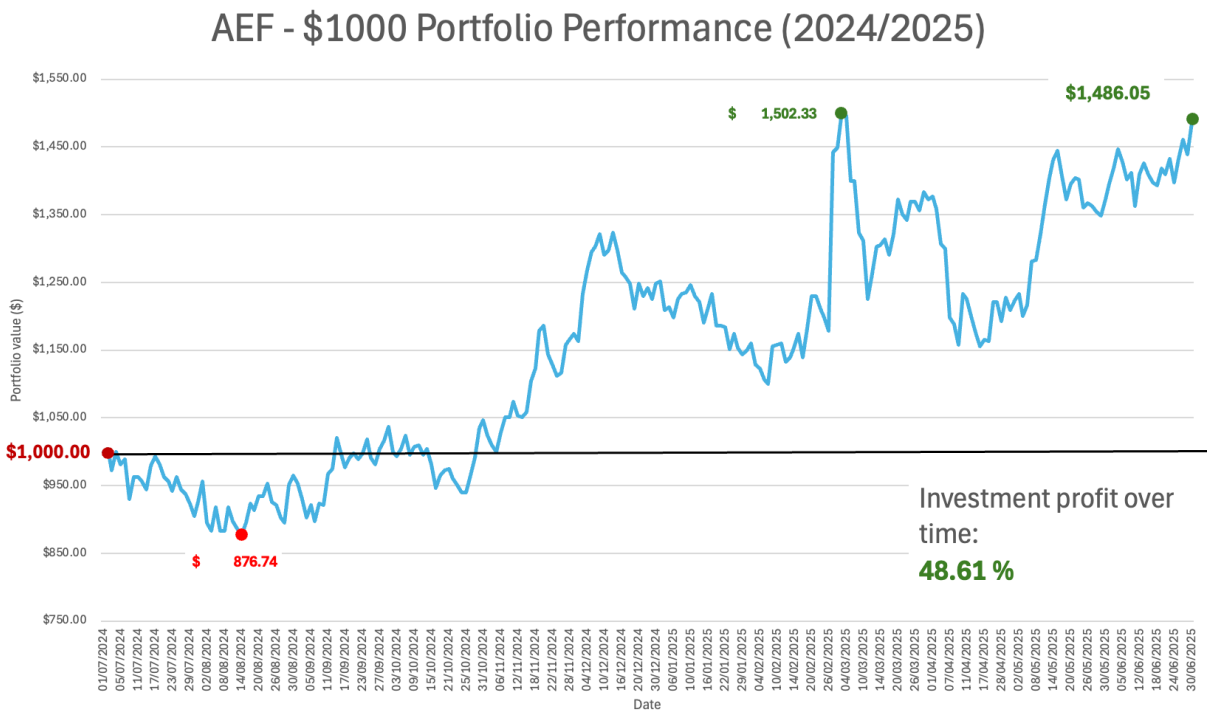


In this percentual cumulative chart, we can spot four main phases. The first one, from July 2024 to October 2024, features a negative yield. The yield starts at 0% and quickly becomes negatives during this period, due to decreases in the close price. The line fluctuates between

0% and -10%. This indicates that during this period, recent investors were facing loss. We can also see a constant intention of the close price trying to increase but doesn't reach it.
In the second phase, between November 2024 and December 2024, the close price break over the 0% line and starts growing rapidly. The line increases quickly, reaching a peak of over 30% cumulative yield over late December. This means a significative profits period for investors.
The third phase, between January 2025 and February 2025, faces a growth stop, and the line remains stable, fluctuating between 20% and 30%. During this period, an investment wouldn't translate into major profits, suggesting that the share price got stabilized, missing a clear growth pattern.
The last phase, from March 2025 to June 2025, shows a yield that starts growing again, reaching a new peak of almost 50%. However, this growth is more volatile than the one on phase two, because it has abrupt drops, but it also recovers and maintain a positive value, between 30% and 45%. The period ends with a solid percentual yield, around 45% and 50%, which suggest that despite volatility, the share has been a profitable investment on the analysed period.

Yearly $1,000 portfolio

For this section, I had to create a new variable called Shares Purchased (column M), to represent the $1,000 bought at the beginning of the period. To calculate this variable, I calculated $1,000 divided by the close price at 1st of July 2024 (beginning of the period), to derive how many shares it is equivalent to, at the time. Then I dragged that number across to column, as the shares purchased won't change across time, what would change, is its value. This is a quantitative ratio-scale variable, represented as number.
Moreover, I created another new variable called Portfolio Value (column N), to track the total value of the investment for each day of the studied period. This amount is obtained by multiplying shares purchased by close price. I dragged this formula across the column, to daily track the portfolio value during the whole studied period. This is a quantitative ratio-scale variable, represented as currency.
Finally, I created a variable called Investment Profit/Loss, calculated by dividing the current portfolio by value minus the initial investment ($1,000), by the initial investment ($1,000), to track the cumulative investment profit or loss over time. This is a quantitative ratio-scale variable, represented as percentage. Positive percentages were labelled in green and negatives in red.
In this chart, I added a referential $1,000 to keep track of the initial invested amount and easily identify profits/losses. I also added label values for $1,000 and the final portfolio value, $1,486.05 to highlight the investment's initial and final values. Lowest and highest portfolio values where highlighted in the graphic too. Finally, I added a text box to highlight the percentual profit of this investment.
This line chart visualizes the performance of a $1,000 hypothetical portfolio invested in AEF over the 2024/2025 financial year.



As an overall trend, we see a line that shows the value of the portfolio rising over the entire year. This trend is generally positive and upward, indicating that the value of the AEF stock increased significantly during this period. The portfolio has experienced a strong capital appreciation.

There is a notable dip in the portfolio's value in the initial months, specifically around August/September 2024, where it seems to have dropped below the initial $1,000 investment. This is the only period where the portfolio experienced loss, because the value quickly recovers and begins a steady, continuous upward climb from late September/early October 2024 onwards.

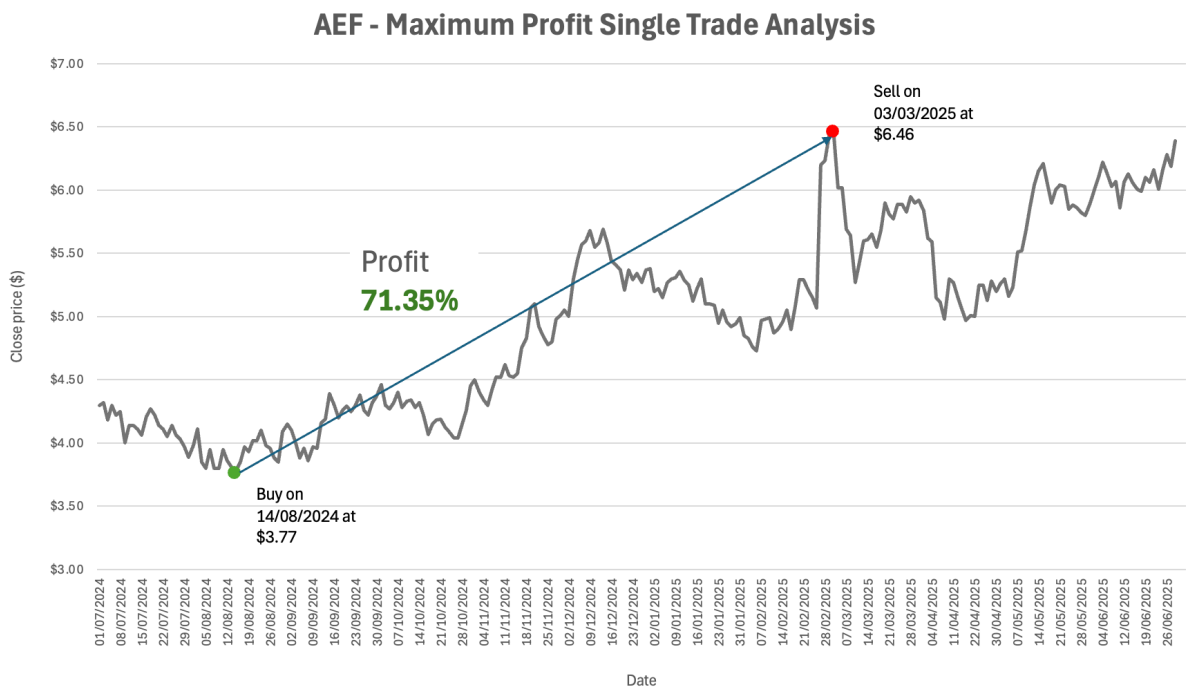
The portfolio appears to reach its peak value towards the end of the financial year, around June 2025, suggesting a positive forecast for the future. The line ends at a value over $1,486, which represents a substantial profit (almost 50%) for the investor who used this buy and hold strategy.

Single trade for maximum profits

For this section, I first identified the best buy price (close price). To do this, I added a section at the right side of the close price column, to identify the highest and lowest price over the studied period, and then highlighted those cells on the column, to link them with their respective date, and relevant information.

Next, I created a chart in the "Single trade for max profits" sheet, showing the number of shares purchased (calculated dividing $1,000 by the lowest price), the portfolio value at sale (calculate multiplying the highest close price by the number of shares purchased), and the total profit (obtained by subtracting the lowest close price from the highest close price, and then multiplying by the number of shares purchased).

In this graph, I marked the lowest close price as the “buy” moment in green, and the highest close price as the “sell” moment in red. Then, I created a line between the buy and sell dates, to visually represent the profit and its amount.

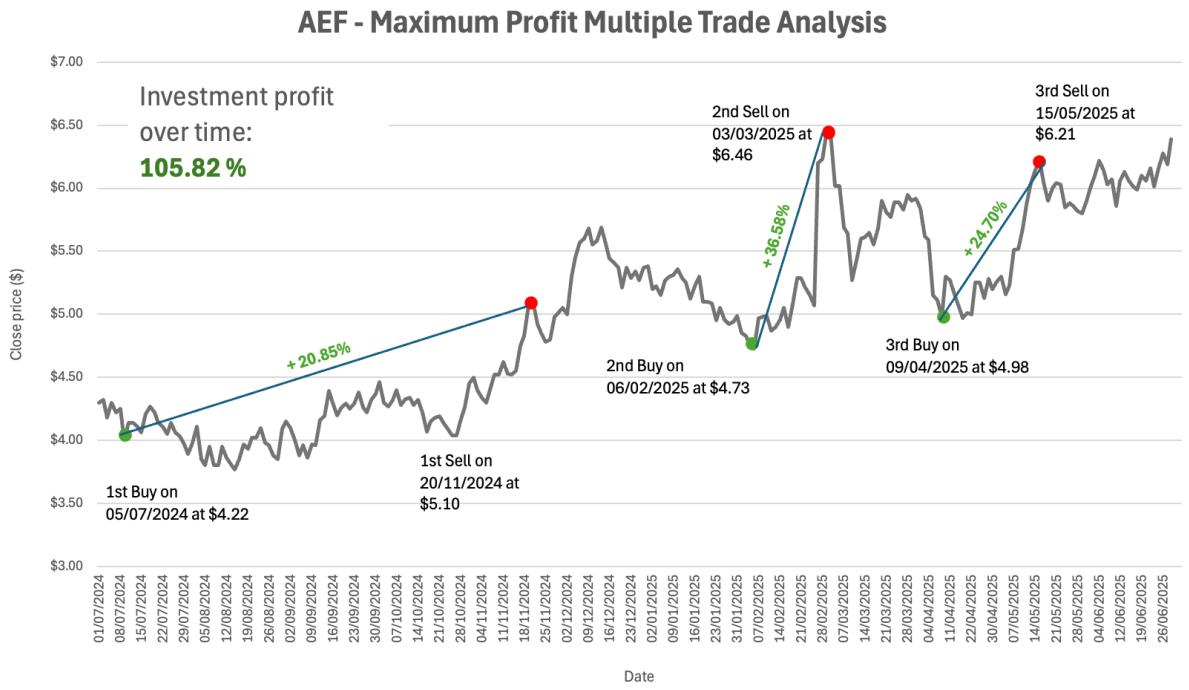


As explained before, I identified the lowest close price ($3.77 on 14/08/2024) and the highest close price ($6.46 on 03/03/2025). The suggested single trade to maximise profits, is to buy on 14/08/2024, when the price is low and 265.25 shares can be acquired with $1,000, and then sell those shares when the close price reach its maximum, on 03/03/2025, when the close price is $6.46. This trade retrieves a gross income of $1,713.53, leading to a profit of $713.53 (by subtracting the initial investment of $1,000), which translates to a 71.35% profit.
While this analysis shows the “best-case scenario”, it also highlights the significant potential for profit from AEF stock’s price volatility. Please note that timing the market perfectly like this is extremely difficult in rea-life trading, which makes the analysis theoretical but still valuable for understanding the stock’s potential.

Multiple trades for maximum profits

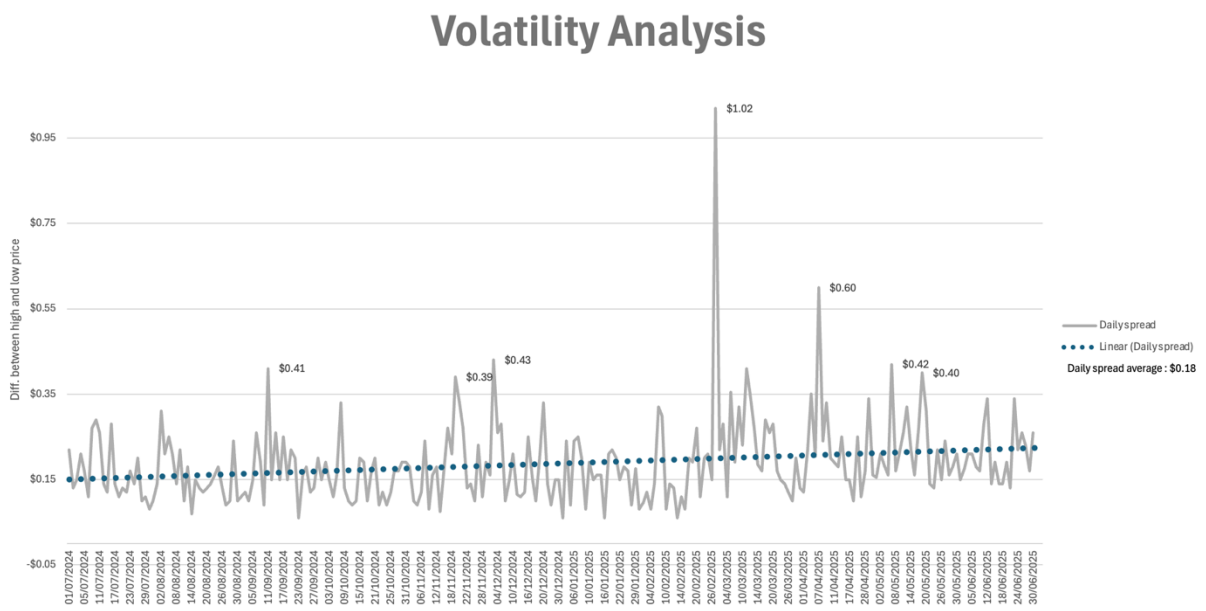
This section was based on three key transactions, from now on, called trades 1, 2 and 3, respectively. I created a table chart in a sheet called “Multi trades for maximum profit”, to further explain this trading strategy.
I suggest executing the first trade (buy) on 05/07/2024, because I wanted to find the first “lower” close value but suggesting a safer investment. The first lower close value, was on 03/07/2024, corresponding to $4.18, but that was a sudden drop from $4.32, so I wanted to find a new lower value after an uptick. On 05/07/2024, the close price is at $4.22. Then, on 20/11/2024, the close price rose significantly from the buy date, reaching a steady peak of $5.10. This is a logical moment to sell, because the stock has experienced an overall upward

and there is a risk of a downward trend. Selling at this point, allows us to lock in the current profit and free up the capital.
The second trade is to buy on 06/02/2025, because after the initial sell-off from the peak experienced in September, the price settled into a new range of prices. On this date, at a price of $4.73, I saw a local low or new “lower” value. This could be considered as a strategic re-entry investment. Now we re-buy stock with the initial $1,000 plus its profits after the first trade, and we obtain a portfolio of 255.5 shares. Following, on 03/03/2025, the price increased, reaching a peak of $6.46. Selling here is a conservative investment idea because it allows us to obtain gain from this second trade before the stock corrects or enters a flat period.
Finally, I suggest buying again on 09/04/2025, at a close price at $4.98 representing a new strategic entry point. This amount could be considered as a local minimum and allows us to reinvest this new fund, obtaining 331.44 shares. Afterwards, we see the final peak of the studied period, with the price hitting $6.21 on 15/05/2025. By selling here, we conclude the financial year locking profits from investment.
In this chart, I added a final column representing profits after each trade, and I added a conditional formatting, to highlight the cell red when the value is equal or less than 0 (loss) and red when is larger than 0 (green). This helps us easily spot profits/losses.



Daily volatility analysis
In this segment, I wanted to visualise the difference between the highest and lowest prices of the stock on each trading day. A large “spread” between those prices, indicates high volatility and uncertainty, while a small spread suggests a more stable and predictable market. However, it is important to have in mind that most of times, higher volatility is tied to higher risks, as well as higher potential profits/losses.

To create a visualisation for this daily volatility, I created a new variable called “Daily spread” (column P), which is obtained by subtracting the low column from the high column, and then dragging down for the whole column.
For this visual, I used a line chart, with date on the x-axis and daily spread on the y-axis. I also added a linear trend-line to show the overall long-term direction. I added data labels for the significant peaks within the chart.
I included a daily spread average label, which is useful to have a reference number of what could be considered as normal spread across this share.



As this chart tells a story about risk, not just return, I identified periods of high volatility, to later analyse possible causes. For the studied period, the daily spread seems to be relatively low and stable, with a few significant spiked and a major one. This suggests that while the stock has been trending upward (as we saw on previous charts), its day-to-day volatility is generally contained, making it a safe investment.
The most notable features on this chart are the sharp peaks. There is a significant spike in volatility around late August/early September 2024, there’s another larger and major peak on 27/02/2025, which is the highest by far. We can also observe smaller peaks around early April 2025 and during May 2025.
These spikes indicate periods where the stock’s price was swinging widely within a single day. This could be a sign of major news (i.e: company announcement, new contract or market-wide event) or just high trading volumes. However, the periods of low volatility (flat sections), indicate times when the stock’s price was very stable, with minimal different between daily highs and lows. This suggests a period of calm or consolidation before bigger movements.

Monthly performance analysis
I created this visualisation to provide a summary of the stock’s yearly performance. The goal of this chart is to display the total percentage gain or loss for the stock in each month of the

2024-2025 financial year. This will allow us to quickly identify the best and worst-performing months, but also to have a deeper understanding of how beneficial an investment in this fund could be.

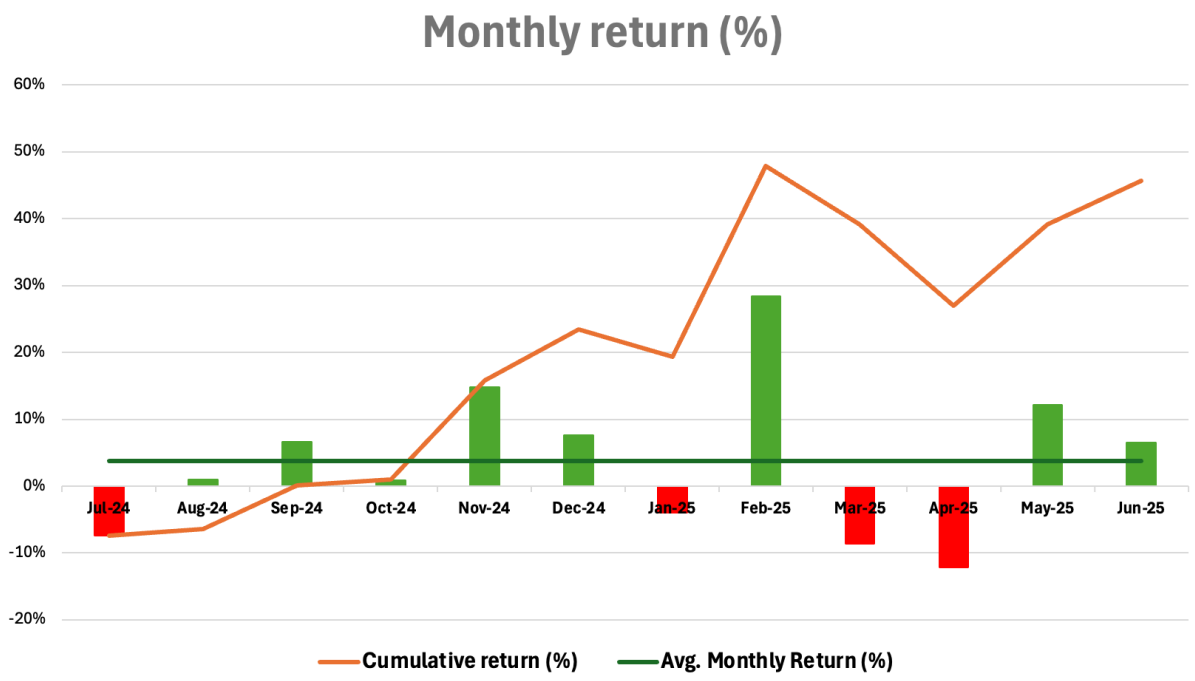
For this chart, I re-arranged the data across 3 new columns, column A contains each month plus its year, column B contains the close price at the first day of each month, during all the studied period and column C contains the close price at the last day of each month. To obtain days, I filtered the date and the close price columns by month and year. Finally, I created a new variable called “Monthly return (%)” in column D, obtained by dividing the close price at the end of the month by the close price at the start of the month and then subtracting 1. I formatted this column to be represented as percentage.

To visualise this, I created a bar chart, using green for all the bars that represent a positive return (above 0%) and red for all the bars that represent a negative return (below 0%).

The reason why I chose to represent monthly performance, is because daily performance leads to too many bars, so it is not that easy to understand the storytelling behind it and doing it quarterly would provide too much information in just one bar, possibly missing relevant information.

I also added a cumulative return line to show the total profit or loss of the stock from the beginning of the year, to illustrate the story of the overall growth over time. I created a new variable (column E), called “Cumulative return (%)”, obtained by summing 1 with the previous month’s cumulative return, divided by 100, and then multiplying by 1 plus the current month’s return divided by 100, and then formatting it as percentage.

Finally, I added a reference line for an average monthly return for the entire period. To obtain this, I added a new row at the bottom of column Monthly Return, that calculated this average. I pasted that value in a new column called “Avg. Monthly Return”, using the same value across all months + year.



From this char, we can see a clear story about the stock's performance over the financial year. First, we observe that the stock experienced a strong start to the financial year, with significant positive returns in August and December 2024. Then, we can infer a notable negative performance in October 2024, which stands out as the month with the largest loss. Subsequently, the stock showed a consistent and positive trend from January to June, with every month ending with a gain. This demonstrates a period of sustainer positive momentum. From the cumulative return line, we can see that it shows consistent overall growth from the start of the period. Despite the large drop in October, the cumulative return line didn't fall below 0, indicating that the initial gains were strong enough to make up the loss. The line also shows a steady, upward trajectory, especially in the second half of the period, ultimately reaching a strong final value of almost 50%.
What we can infer from this, is that the monthly performance demonstrates that despite minor monthly fluctuation, the stock has been a strong long-term investment and show signs of remaining as a good investment.

Summary

Based on the comprehensive analysis of Australian Ethical Investment Ltd. (AEF), I could do the following recommendations for potential investors.
First, the most important finding is that there is a strong upward trend in the stock's price, specially from late 2024 to mid-2025 (the second part of the studied period). By performing a simple buy-and-hold strategy with a $1,000 investment could have resulted in a profit of almost 50%. Also, the monthly performance chart supports this as the cumulative return line did not fall below zero from the beginning. This shows that the company's value is consistent and gradually appreciated, indicating a strong investment.
However, every investment carries a risk with it. For investors with a higher risk tolerance, this stock presents opportunities for strategic trading to maximise profits. The single-trade analysis showed large profits and furthermore, a multiple-trade strategy could have taken advantage of local peaks and troughs, generating an even higher profit of around 100%.
On the other hand, while the stock's overall day-to-day volatility is relatively low and stable, it has sharp spikes, but they could be linked to periods of significant stress and price movement. Investors should monitor these periods closely.
Finally, the correlation analysis at the beginning of this report, revealed that the relationship between trading volume and the stock's price if very weak (around 0.2), suggesting that volume alone is not a primary driver of the stock's price movements and should be considered with other factors, such as company announcements or market events. In contrast, the analysis of shares on issue vs. market capitalisation showed a strong positive correlation, indicating that the act of issuing shares, was interpreted as a positive signal by the market, leading to an increase in the stock price.

Even though AI tools can be very helpful in analysing data and drive to conclusions, the visualisation approaches in this report, provided a clear narrative that is difficult to be provided by AI, as each chart was created by hand on Excel.

In this report, I created visualisations that transformed raw data into visual storytelling. For example, using combo charts with secondary axes and reference lines, may not be the first approach that AI would suggest. Also, these visualisations facilitated the interpretation of key insights that could be missed in an AI analysis.

On the other hand, using different chart types was crucial to effectively illustrate specific relationships. The capital vs issued shares analysis used a line chart with a secondary axis to visually provide the correlation between these variables.

Finally, this report used simple techniques like colour coding and labelling to provide further information. For instance, making profit/loss in green/red made it easier to spot the results of each transaction in the multiple trades for maximum profit chart. Turning complex data into easily and understandable information is a human-centric approach.